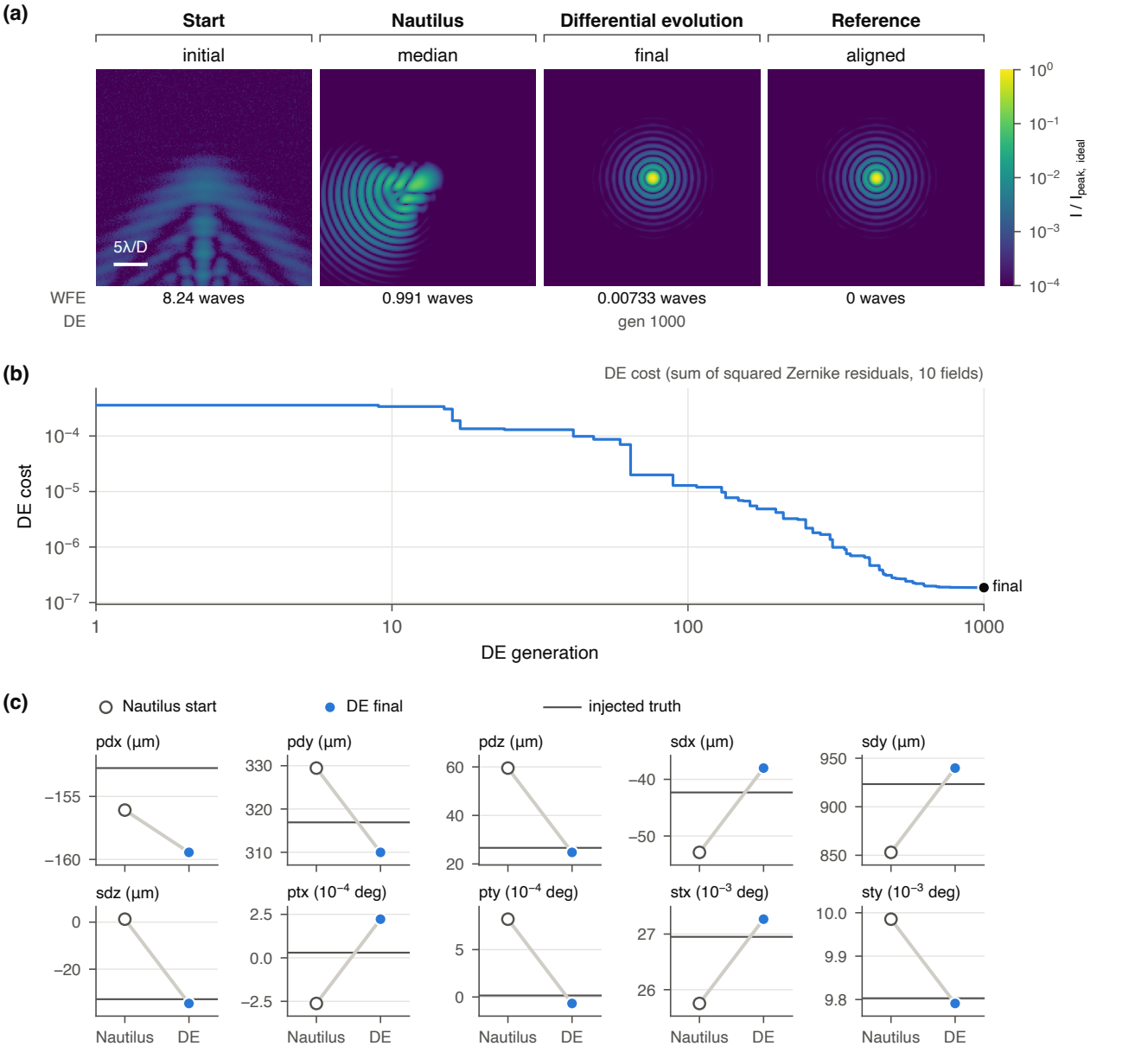

(a)
Start
Nautilus
Differential evolution
Reference
initial
median
final
aligned
5λ/D
WFE
DE
8.24 waves
0.991 waves
0.00733 waves
0 waves
gen 1000
I / Ipeak, ideal
(b)
DE cost (sum of squared Zernike residuals, 10 fields)
DE cost
DE generation
final
(c)
Nautilus start
DE final
injected truth
pdx (μm)
pdy (μm)
pdz (μm)
sdx (μm)
sdy (μm)
sdz (μm)
ptx (10⁻⁴ deg)
pty (10⁻⁴ deg)
stx (10⁻³ deg)
sty (10⁻³ deg)
Nautilus
DE